

Cardiovascular Mortality Over Time: Data-driven exploration using public WHO mortality data

Carmine Scialdone (ID6436684)

Date: October 14, 2026

Introduction

Cardiovascular diseases are the leading cause of death worldwide. Studying the countries with the highest cardiovascular mortality rates, and how these rates change over time, can be an important starting point for analyzing differences between populations, and understanding whether these differences may be influenced by prevention strategies, risk factors, or characteristics of healthcare systems. In this report we focus on a simple research question: **Which five countries reported the highest cardiovascular disease death rates in 2020, and how did some of these rates change annually between 2000 and 2020?**

To answer this question, we analyzed a public dataset available on Our World in Data, based on data from the WHO Mortality Database.

Data sample

The dataset contains more than 4,900 total observations, which come from the fact that, for each of the 117 countries represented in the dataset, data are available for several different years.

The relevant columns used in this analysis include:

- **Entity**: The country or the location considered.
- **Code**: Country code.
- **Year**: Year of the observation.
- **Death rate**: The variable "Age-standardized deaths that are from cardiovascular diseases per 100,000 people, in both sexes aged all ages", simplified in the analysis as "death rate".

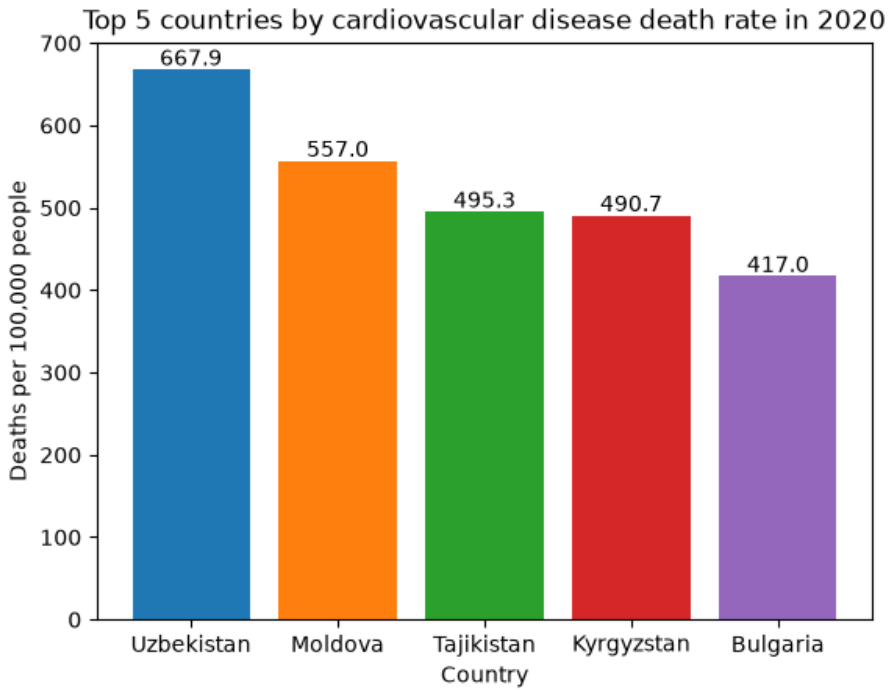


Figure 1: Top five countries with the highest cardiovascular disease mortality rates in 2020.

The dataset [1]...

Results and discussion

The first observations allowed us to identify the five countries with the highest age-standardized cardiovascular disease death rates per 100,000 people, considering both sexes and all ages: Uzbekistan, Moldova, Tajikistan, Kyrgyzstan, and Bulgaria.

To better understand how high these values are compared with the overall distribution, we also analyzed the cardiovascular mortality rates of all countries available (77 countries) in 2020 using descriptive statistics.

$$\text{Mean} \approx 179.8 \pm 23.3 \text{ (subsampling-based estimate)}$$
$$\text{Standard deviation} \approx 121.6$$
$$\text{Skewness} \approx 1.73$$

The values obtained indicate different behaviors. The mean cardiovascular mortality rate suggests that, on average, the mortality rate is much lower compared with the five countries analyzed. The standard deviation, which is quite large compared with the mean value, suggests that mortality rates vary considerably from country to country. The last result, the skewness, suggests that, since it is positive, the distribution extends towards higher values, with a smaller number of countries showing particularly high mortality rates.

The following graph allows us to examine the trend in mortality rates over the twenty-year period preceding 2020. Uzbekistan and Tajikistan were not included due to a lack of available data.

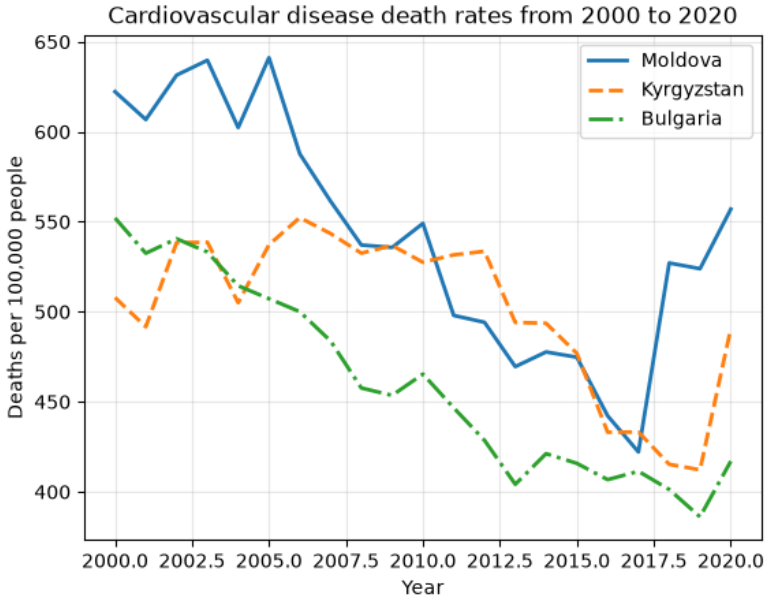


Figure 2: Variation in cardiovascular disease mortality rates over the 2000–2020 period.

For each country, the annual percentage change was calculated by comparing each year with the previous one: $P_t = \frac{R_t - R_{t-1}}{R_{t-1}} \times 100$. The mean annual percentage change was then calculated as: $\overline{P} = \frac{1}{n} \sum_{i=1}^{n} P_i$.

Results: Moldova $-0.31 \pm 2.32\%$, Kyrgyzstan $0.00 \pm 1.01\%$, and Bulgaria $-1.34 \pm 0.84\%$ (SD of 3 subsample means). The most interesting result is Kyrgyzstan, where the average change is close to zero, not because the rate remained stable, but because annual increases and decreases compensated each other.

References

[1] WHO Mortality Database (2025), with minor processing by Our World in Data. *Death rate from cardiovascular diseases*. Available at: https://ourworldindata.org/grapher/cardiovascular-disease-death-rate-who-mdb.